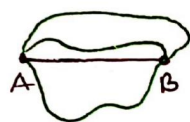


Distance & Displacement:

$$|\text{Displacement}| \leq \text{Distance}$$

① For any two points



→ only one possible displacement

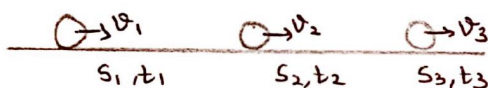
→ Infinite distance possible

Speed & Velocity:

(Average)

$$\langle \vec{v} \rangle = \frac{\text{Total displacement}}{\text{Total time}}$$

$$\langle \text{speed} \rangle = \frac{\text{Total distance}}{\text{Total time}}$$



① When $t_1 = t_2 = t_3$

$$\langle \vec{v} \rangle = \frac{v_1 + v_2 + v_3}{3}$$

② When $s_1 = s_2 = s_3$

$$\langle \vec{v} \rangle = \frac{3v_1 v_2 v_3}{v_1 v_2 + v_2 v_3 + v_1 v_3}$$

→ If particle move with same direction in st. line, then

$$\text{Distance} = |\text{Displacement}|$$

$$\text{Speed} = |\text{Vavg}|$$

(Instantaneous)

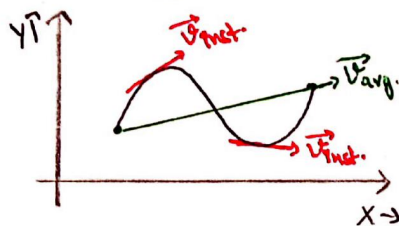
$$\vec{v} = \frac{dx}{dt}$$

Slope (tangent) of x-t curve

$$|\vec{v}| = \text{speed inst.}$$

Direction of inst. velocity is tangential to path while that of avg. velocity is along displacement.

Motion in 1-D



Acceleration:

$$\langle \vec{a} \rangle = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2x}{dt^2} = v \cdot \frac{d\vec{v}}{dx}$$

Constant Acceleration $\Rightarrow$

$$\vec{v} = \vec{u} + \vec{a}t$$

$$(\vec{v})^2 - (\vec{u})^2 = 2\vec{a} \cdot \vec{s}$$

$$\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2 = \left(\frac{\vec{v} + \vec{u}}{2}\right)t$$

Retardation $\Rightarrow$

$$t = \frac{u}{a}$$

$$s = \frac{u^2}{2a} \quad (\text{stopping distance})$$

Displacement in nth sec. $\Rightarrow$

→ nth sec. is from (n-1) sec to n sec.
→ Duration of every nth sec is 1 sec.

$$s_{n\text{th}} = ut + \frac{1}{2}a(2n-1)$$

→ If $u=0$, then

$$\text{Displacement in } n \text{ sec} \Rightarrow s_1 : s_2 : s_3 : \dots \\ (1)^2 : (2)^2 : (3)^2 : \dots$$

$$\text{Displacement in } n^{\text{th}} \text{ sec} \Rightarrow s_{1\text{st}} : s_{2\text{nd}} : s_{3\text{rd}} : \dots \\ 1 : 3 : 5 : \dots$$

Motion Under Gravity:

(I) Projection from ground $\Rightarrow$

$$\uparrow \text{ +ve} \quad \downarrow \text{ -ve} \quad (i) t_{\text{ascent}} = t_{\text{descent}} = \frac{u}{g}$$

$$(ii) T = \frac{2u}{g}$$

$$(iii) H = \frac{u^2}{2g}$$

$$(iv) \text{Velocity at } \frac{H}{2} \Rightarrow \frac{u^2}{2} = v^2$$

(*) Collection of ∞ projection from diff. heights

(*) Motion is symmetrical about maxima

(II) Projection from Height $\Rightarrow$

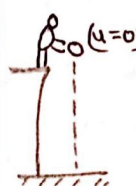
$$(i) T = t_{A \rightarrow B} + t_{B \rightarrow C}$$

$$T = \frac{u}{g} + \sqrt{\frac{u^2}{g^2} + \frac{2H}{g}}$$



$$(ii) H_{\text{max}} = \frac{u^2}{2g} + H$$

(III) Body dropped from Height $\Rightarrow$



$$(i) T = \sqrt{\frac{2H}{g}}$$

$$(ii) v_{\text{strike}} = -\sqrt{2gH}$$

Air Drag: Acts opp. to velocity of body



$$\vec{F} = -b\vec{v}$$

Moving Body: Body thrown/released from a balloon moving up, its velocity will be net of v_{balloon} & v_{throw} .



$$\vec{v}_{\text{ball}} = +22 \text{ m/s}$$

(IV) Body Thrown From Height $\Rightarrow$



$$T = \sqrt{\frac{u^2}{g^2} + \frac{2H}{g}} - \frac{u}{g}$$

Graphs:

→ Tangent at the point, gives slope at that point.

→ Physical significance of slope $\Rightarrow$

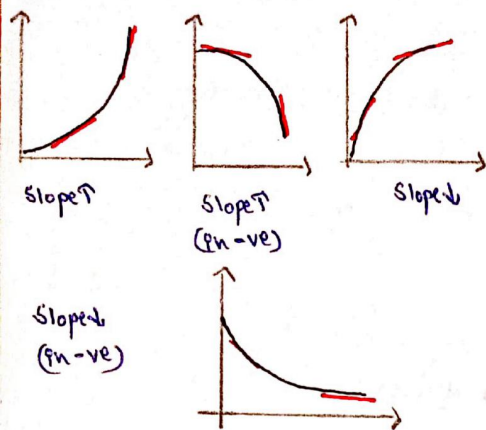
(i) If, one parameter $\uparrow$ Another parameter $\uparrow \Rightarrow$ +ve slope

(ii) If, one parameter $\uparrow$ Another parameter $\downarrow \Rightarrow$ -ve slope

(iii) If, one parameter changes Another parameter const. $\Rightarrow$ zero slope

→ At $\theta = 90^\circ$, no slope defined, no relation b/w given parameters.

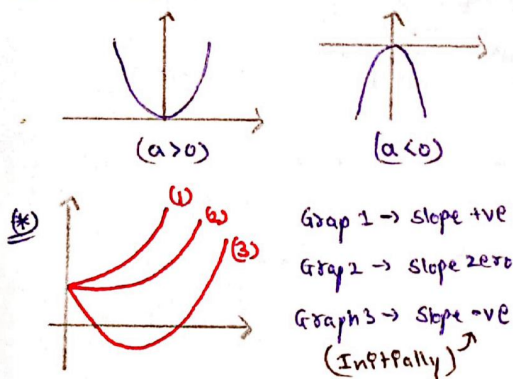
Variation of Slope $\Rightarrow$



$\rightarrow$ Area under $v-t$ curve $\Rightarrow$ Displacement

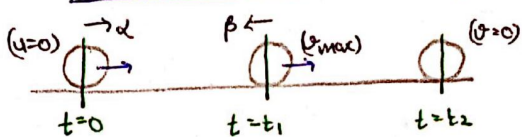
$\rightarrow$ Area under $a-t$ curve $\Rightarrow$ Velocity change

Parabolic Graph $\Rightarrow$



Most Asked Question:

Retardation & Acceleration together

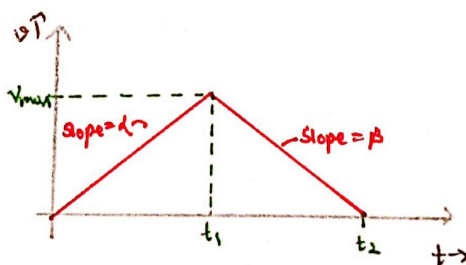


Here, $\alpha \rightarrow$ Acceleration for t_1
 $\beta \rightarrow$ Retardation for t_2

$$(i) \quad t_1 = \left(\frac{\alpha}{\alpha + \beta} \right) T \quad t_2 = \left(\frac{\beta}{\alpha + \beta} \right) T$$

$$(ii) \quad v_{\max} = \alpha t_1 = \beta t_2 = \left(\frac{\alpha \beta}{\alpha + \beta} \right) T$$

$$(iii) \quad S = \frac{1}{2} \left(\frac{\alpha \beta}{\alpha + \beta} \right) T \quad (iv) \quad \frac{t_1}{t_2} = \frac{\beta}{\alpha}$$



Acceleration-Displacement Graph $\Rightarrow$

Here, use $a = v \cdot \frac{dv}{dx}$

$$\int_{x_1}^{x_2} a \cdot dx = \int_{v_1}^{v_2} v \cdot dv$$

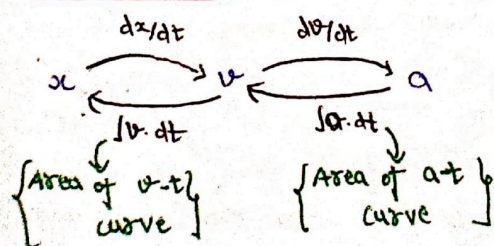
Velocity-Displacement Graph $\Rightarrow$

(i) Write the equation of $v = f(x)$

(ii) Break into differential form & Integrate.

Simplest approach for accn-velocity graph

Variable Acceleration:



Vectors:

(i) Unit vector: $\hat{A} = \frac{\vec{A}}{|\vec{A}|}$

(ii) We can write any vector as the sum of its mutually per components

$$\vec{A} = x\hat{i} + y\hat{j} + z\hat{k}$$

(iii) Angle between two vectors is measured by tail-to-tail joining.



(iv) We can break any vector into its components as -



$$|\vec{v}| = \sqrt{(v \cos \theta)^2 + (v \sin \theta)^2}$$

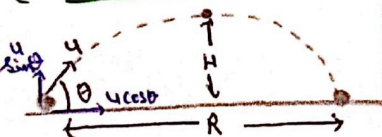
(v) Two mutually per defn are independent of each other.

Motion of x → Governed by a_x, v_x

Motion of y → Governed by a_y, v_y

Projectile Motion:

(I) Ground to Ground ⇒



(Assume)

- Flat Ground
- Const. g
- No air drag

(i) $T = \frac{2u_y}{a_y} = \frac{2u \sin \theta}{g}$

(ii) $t_{asc} = \frac{u_y}{g} = \frac{u \sin \theta}{g}$

(iii) $H = \frac{(u_y)^2}{2a_y} = \frac{u^2 \sin^2 \theta}{2g}$

(iv) $R = \frac{u^2 \sin 2\theta}{g}$

For H_{max} , $\theta = 90^\circ$

For R_{max} , $\theta = 45^\circ$

If two same range, then

$$R_1 = R_2 \Leftrightarrow \theta_1 + \theta_2 = 90^\circ$$

(Complementary Angles)



Motion in 2-D

① T, H and u_y are proportional to each other

$$\therefore u_y T \Rightarrow T \uparrow \Rightarrow H \uparrow$$

② When $a_x = 0$, then $u_x = \text{const.}$ through out motion.

③ $u_y = 0$ at topmost point, also motion is symmetrical about per through topmost.

④ When $a_x = 0$, $v_{avg} = u \cos \theta = u_x$

(II) Projection from Height ⇒

1) Horizontal Projection ⇒



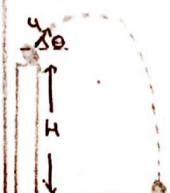
(i) $T = \sqrt{\frac{2H}{g}}$

(ii) $R = u \sqrt{\frac{2H}{g}}$

(iii) $V_{strike} = -\sqrt{2gH}$

(iv) $\vec{V}_t = u_x \hat{i} - g t \hat{j}$

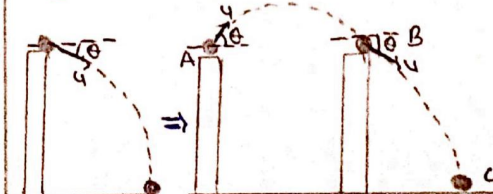
2) Upward Projection ⇒



(i) $T = \frac{u_y}{g} + \sqrt{\frac{u_y^2}{g^2} + \frac{2H}{g}}$

(ii) $R = u_x \cdot T$

3) Downward Projection ⇒



$$T = \sqrt{\frac{u_y^2}{g^2} + \frac{2H}{g}} - \frac{u_y}{g}$$

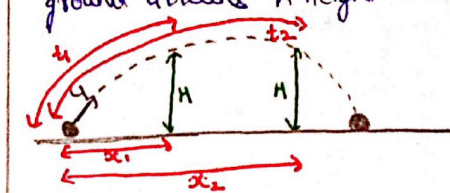
Equation of Trajectory:

$$y = x \tan \theta - \frac{1}{2} \frac{g x^2}{u_x^2}$$

where, $u_x = u \cos \theta$

$$\tan \theta = \frac{u_y}{u_x} = \frac{u \sin \theta}{u \cos \theta}$$

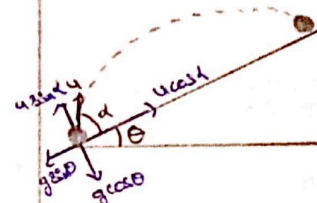
→ A particle projected from height ground attains H-height twice.



Here, $x_1 + x_2 = R$ & $t_1 + t_2 = T$

Projectile on Incline:

(I) Upward Projection ⇒



(i) $T = \frac{2u \sin \alpha}{g \cos \theta}$

(ii) $H = \frac{u^2 \sin^2 \alpha}{2g \cos \theta}$

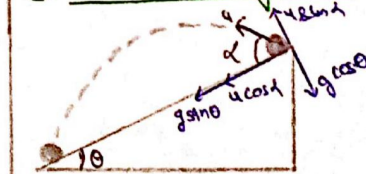
(iii) $R = \frac{x}{\cos \theta} = \frac{u \cos (\alpha + \theta) \cdot T}{\cos \theta}$

$$R = \frac{2u^2 \sin \alpha \cdot \cos (\alpha + \theta)}{g \cos^2 \theta}$$

For R_{max}

$$\alpha' = \frac{90^\circ - \theta}{2} \Rightarrow R_{max} = \frac{u^2}{(1 + \sin \theta) g}$$

(II) Downward Projection ⇒



(i) $T = \frac{2u \sin \alpha}{g \cos \theta}$

(ii) $H = \frac{u^2 \sin^2 \alpha}{2g \cos \theta}$

(iii) $R = \frac{2u^2 \sin \alpha \cdot \cos (\alpha + \theta)}{g \cos^2 \theta}$

$$R_{max} \text{ at } \alpha = \frac{90^\circ + \theta}{2}$$

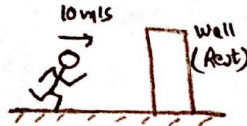
Condition of per incidence ⇒

$$2 \tan \theta = \cot \alpha$$

Condition of Horizontal incidence ⇒

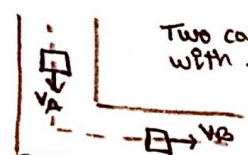
$$2u \sin \alpha = g \sin (\alpha + \theta) \cdot \cos \theta$$

Relative Velocity Concept :

1)  $\vec{V}_{A/B} = \vec{V}_A - \vec{V}_B$
 $\vec{V}_A$ & $\vec{V}_B$ are velocities in ground frame.

→ So here, velocity of wall w.r.t man can be written as,

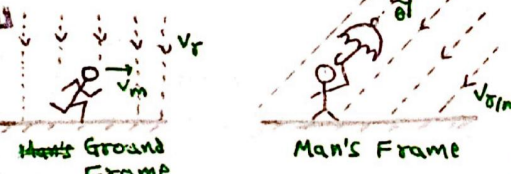
$$\begin{aligned}\vec{V}_{w/m} &= \vec{V}_w - \vec{V}_m \\ &= 0 - (10\hat{i}) \\ &= -10\hat{i} \text{ m/s}\end{aligned}$$

2)  Two cars moving with two velocities.

From car B's frame

$$\vec{V}_{A/B} = \sqrt{V_A^2 + V_B^2} (\hat{i} + \hat{j})$$

Rain Man Problem :

Q1) 
 Ground Frame Man's Frame

→ In order to find $\vec{V}_{r/m}$ " $\vec{V}_r$ vector mein man ki velocity $\vec{V}_m$ par se ke vectorially add kardo."

(i) $\vec{V}_{r/m} = \vec{V}_r - \vec{V}_m$
 (ii) $\vec{V}_{r/m} = \sqrt{V_m^2 + V_r^2}$
 (iii) $\tan\theta = \frac{V_m}{V_r}$

As velocity of man, V_m ↑ ses, θ ↑ ses

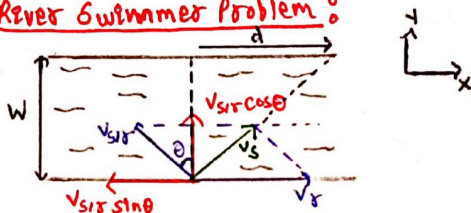
→ Another approach of ques solving,

$$\begin{aligned}\vec{V}_{r/m} &= \vec{V}_r - \vec{V}_m \\ \Rightarrow \vec{V}_r &= \vec{V}_m + \vec{V}_{r/m}\end{aligned}$$

Relative Motion

→ Man jis der'n mein bhagta hai, rain uske opp. der'n mein rotate hone lagti hai.

River Swimmer Problem :



$\vec{V}_r$: Velocity of river w.r.t ground

$\vec{V}_{sr}$: Velocity of swimmer w.r.t river (Velocity in still water)

$\vec{V}_s$: Velocity of swimmer w.r.t ground

d : Drift (Displacement along x-axis)

$V_{sr} \cos\theta$ → Responsible for crossing of river

$$t = \frac{W}{V_{sr} \cos\theta}$$

$(V_r - V_{sr} \sin\theta)$ → Responsible for drift

$$d = (V_r - V_{sr} \sin\theta) t$$

3) Cross River in Minimum Time →

Keep $\vec{V}_{sr}$ ⊥ to $\vec{V}_r$.

$$t_{min} = \frac{W}{V_{sr}}$$

$$\tan\phi = \frac{V_{sr}}{V_r}$$

4) Shortest Path / Zero Drift →

$$V_r \leq V_{sr} \Rightarrow \sin\theta = \frac{V_r}{V_{sr}}$$

5) Minimum drift (d) →

When, $V_r > V_{sr}$, d can be zero but it can be minimum when,

$$\sin\theta = \frac{V_{sr}}{V_r}$$